Report of:

The Freshwater Fish Diversity around Mesangat watershed, District Muara Ancalong, Regency Kutai Kartanegara, Province Kalimantan Timur

by:
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Introduction

REA KON, The conservation section of PT REA Kaltim Plantations need to gather the aquatic fauna baseline data from the concessions area of PT REA KALTIM PLANTATION. Two survey conducted in Ulu Belayan river streams, Mahakam river drainage, District Kembang Janggut, Regency Kutai Timur, Province East Kalimantan. This third survey studied the freshwater fish diversity around Mesangat watershed, District Muara Ancalong, Regency Kutai Kartanegara, Province Kalimantan Timur.

There is a quite big swampy area in the District Muara Ancalong, Mesangat swamp or in Bahasa Indonesia we call it Rawa Mesangat. This swamp area is the habitat of the protected species of long snout crocodile, *Tomistoma schlegeli.*

The aim of this survey is to get the information of the fish diversity around Mesangat watershed, the distribution of each site and the status of the species. The results of this survey could be use as the basic data for REA KON to manage the area for the continuation and conservation of the species.

Material and Methods

The survey obtained between 4–11 November 2009. Fish collected by using electrofishing, tray net, scoup net and some trapped fish nets: gill net, 'anco' and 'buu'. The fish collected from 10 sites:

- Mesangat Ulu river, the out let of Mesangat swamp, flows to Kelinjau river
- 6 tributaries of Mesangat river:
  - Lakum 1
  - Lakum 2
  - Lakum 3
  - Loa Sangkal
  - Loa Beloma
  - Loa Toh
- Mesangat swamp
- Long Ngetek, a small stream flos to Kelinjau river
- Mesangat Ilir, outlet of Mesangat swamp flows to Wahau river

The fish specimens fixation, preservation and procedures are as same as the previous survey. The fish identification based on Brittan (1954), Hadiaty & Siebert (1998); Hadiaty & Siebert (2001)**;** Inger & Chin. (1962); Karnasuta (1993), Kottelat (1984; 1990; 1994; 1995; 1999), Kottelat et al. (1993), Kottelat & Lim (1993), Kottelat & Whitten (1996), Nelson (1994); Ng (1999), Ng & Ferraris (2000), Roberts (1989,1993), Tan & Kottelat (1998), Tan & Ng (2006), Weber & de Beaufort (1916; 1965).

All of the fish specimens deposited in Ichthyology Laboratory, Division of Zoology, Research Center for Biology , Indonesian Institute of Sciences (LIPI, formerly the Museum Zoologicum Bogoriense, MZB). The specimen registered in 176 lots of MZB catalog number, from MZB 18048 -18224.

Results and discussions

The survey in the Muara Ancalong waters found the fact: the fish population in Muara Ancalong waters are still in good condition, especially around Mesangat rivers. There are many fishermen from Muara Ancalong come to Mesangat river for fishing everyday. Some of them have a small floating house at Mesangat river which it call 'rakit'. They usually use gillnet, 'rawai' (a long line with many hooks on it) or 'buu' (a fish trap made from bamboo) etc., but some of them use an electroschoker (5 parallel motorcycle batteries) with a long bamboo stik.

The base camp of REA KON is the second rakit from the mouth of the Mesangat river, every boat pass the camp. The fishermen were very nice and helpful, when they knew we look for the fish diversity in the area, they provided us the fish specimen of the species which we need. We collected around the base camp, by using the scoup net and electrofishing and picked up some from the fishermen, the most effective way to collect the fishes. No wonder, we got about 27 species more than half of the total species at the first day.

The survey successfully collected 1424 fish specimens, consisted of 43 species of 18 familia (Table 1). One of the collected species is the protected animal in Indonesia law: PP No. 7 Tahun 1999. The species is *Notopterus notopterus* (Pallas, 1769), local people named it ikan Belido (Fig. 1, left).

Besides collected the fish species, we also gathered the secondary data by showing the fish guide book to the fishermen. By this activity we got the information of the fish diversity in the area, also the local name of each species (Table 2). From the list there might be 21 species which are not collected in this survey. So, the fish diversity of the Mesangat area supposedly around 64 fish species.

The fishermen usually look for the snake heads: *Channa lucius, C. micropeltes* and *C. striata*, but principally they collect all the fish trapped in the nets, i.e the kissing gouramy, *Helostoma temmincki, Notopterus notopterus, Barbonymus* sp, *Osteochilus* spp etc. Most of them come early in the morning, checking their trapped nets, and go back to Kelinjau village at the afternoon, they sell the fish, if not sold out they make salty fish.

When we go to the Mesangat swamp, there were many dead fish trapped at the gillnets, Pak Yus, the owner of the gillnet said, he didn't check it for three days! Some *Notopterus notopterus,* big kissing gourami, *H. temmincki* etc. The fish dead because the gillnet trapped the gill, the fish can't go or move. But from the other fish trap, 'buu', most of the fish trapped in this tool are still alive.

Table 1. Fish diversity around Mesangat watershed, District Muara Ancalong, Regency Kutai Kartanegara, Province Kalimantan Timur

No	Familia	No	Species	Msgt Ulu	Lakum 1	Lakum2	Lakum 3	Swamp	L Sangkal	L Beloma	L Toh	L Ngetek	Msgt Ilir	TOTAL
				6 -9 Oct	6-Oct	7-Oct	7-Oct	7-Oct	8-Oct	8-Oct	8-Oct	9-Oct	9-Oct	
1	Notopteridae	1	*Notopterus notopterus*	2				1						3
2	Anguillidae	2	*Anguilla cf bicolor*								1			1
3	Cyprinidae	3	*Barbonymus* cf. *gonionotus*										1	1
		4	*Barbichthys laevis*	1										1
		5	*Cyclocheilichthys armatus*	25	1		2		3			1	1	33
		6	*Cyclocheilichthys apogon*	7										7
		7	*Labiobarbus* cf. *kuhlii*	19	3				2	1	3			28
		8	*Leptobarbus hoevenii*	9		1								10
		9	*Osteochilus hasseltii*	23	23			1		4	8	1	1	61
		10	*O.* cf. *microcephalus*				5		10	17	3			35
		11	*Oxygaster* cf. *anomalura*	29	1	1	7	3	3	1	2	3		50
		12	*Parachela hypophthalmus*	5				1	1				4	11
		13	*Puntius binotatus*	1										1
		14	*Puntius tetrazona*	13	17	3	2		13	5				53
		15	*Puntioplites waandersi*	5										5
		16	*Puntius* sp.	1	1		2		3		1			8
		17	*Rasbora cf. hubbsii*	113	58	2	15	199	14		42	43	55	541
		18	*Rasbora* sp	12	14			186						212
		19	*Thynnichthys cf thynnoides*	8		6					3		3	20
4	Balitoridae	20	*Homaloptera* sp juv.								1			1
5	Nemacheilidae	21	*Nemacheilus spiniferus*								1			1
6	Cobitidae	22	*Kottelatlimia* sp				1			1		4		6
		23	*Pangio* sp.				2		8		32	11		53
7	Bagridae	24	*Hemibagrus nemurus*	1	1									2
		25	*Mystus micracanthus*	4	3				10	1	2			20
		26	*Mystus* cf. *nigriceps*	10					2		2	1		15
8	Siluridae	27	*Ompok eugeneiatus*	25	30		4			1		18	3	81
		28	*Ompok bimaculatus*	5		1	1		2	2				11
9	Clariidae	29	*Clarias nieuhofi*	1										1

10	Belonidae	30	*Xenentodon cancilloides*	6										6
11	Hemirhamphidae	31	*Hemirhamphodon* sp.	8	9	2	6	1	10	27		2		65
12	Mastacembelidae	32	*Macrognathus* sp	2										2
13	Pristolepididae	33	*Pristolepis fasciata*	5					1					6
14	Eleotrididae	34	*Oxyeleotris marmorata*	1										1
15	Helostomatidae	35	*Helostoma temmincki*	4										4
16	Anabantidae	36	*Anabas testudineus*	1										1
17	Belontiidae	37	*Trichogaster trichopterus*					2			1			3
		38	*Trichogaster pectoralis*	1										1
		39	*Trichopsis vittata*	3	17	2	1		2	7				32
		40	*Betta sp*		4	1	6		11		3			25
18	Channidae	41	*Channa lucius*	1	2									3
		42	*C. striata*	2										2
		43	*C. micropeltes*			1								1
				353	184	20	54	394	95	67	105	84	68	1424

Figure 1. The fish collected from Mesangat waters: left, *Notopterus notopterus* (Pallas, 1769), one of the protected fish in Indonesia (PP No.7 Tahun 1999); right, the snake heads, *Channa* spp. the main fish catchment from this area.

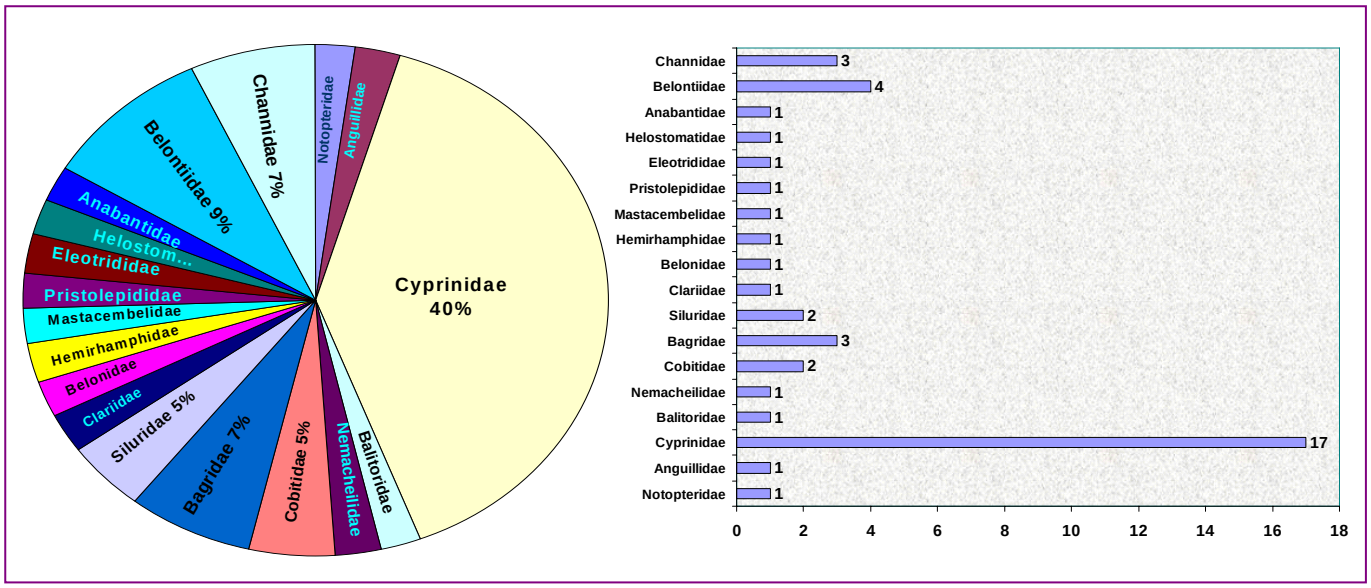
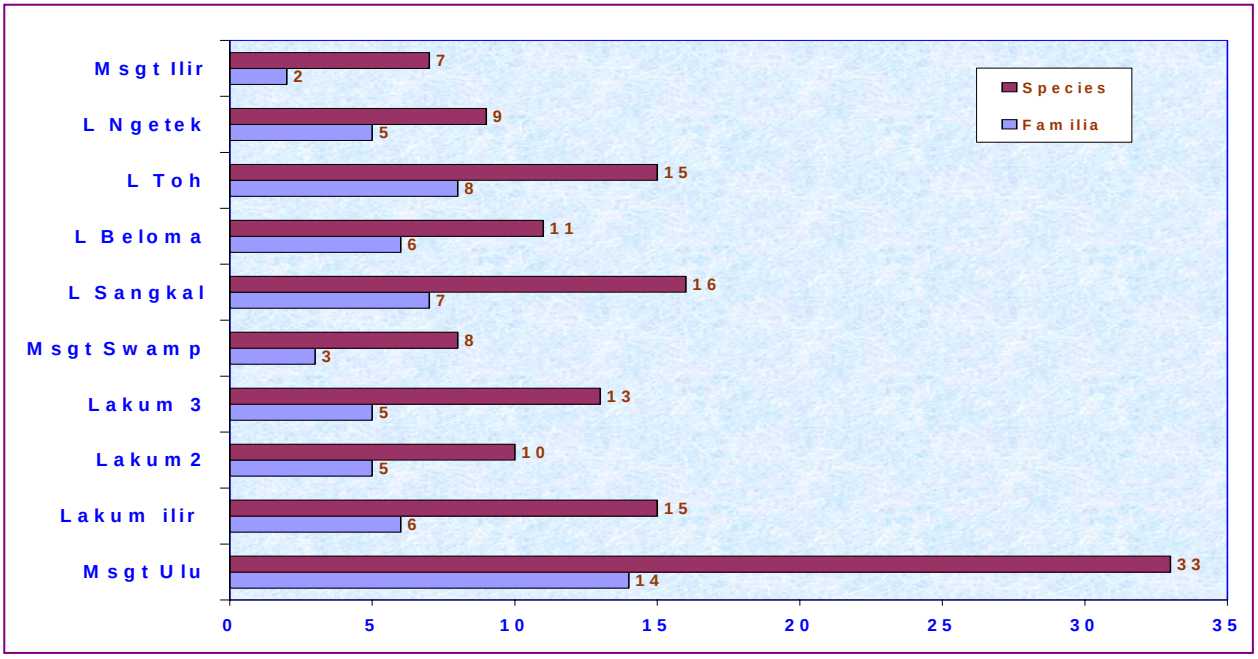


Figure 2. Percentage of the familia (left) and number of species of fish collection of each familia (right) in Mesangat watershed

Cyprinidae is the dominant family of the fish collection, it consisted of 40 % or 17 species. The other familia i.e Belontiidae, Bagridae, Channidae and Cobitidae was just lest than 10 species or less than 10 %.



The fish specimens collected from 10 sites. The highest number of familia and species collected at Mesangat Ulu river (14 familia and 33 species). Mesangat Ulu is the main river in the Mesangat watershed, the other 7 sites are the tributaries of this river (Loa Lakum ilir, L Lakum 2, Lakum 3, Mesangat swamp, L Sangkal, L Beloma and L Toh). The familia found in the tributaries were less than half of the familia in the main river, also the number of the species.

Tabel 2, The secondary data of the fish diversity around Mesangat river and the local name of each fish species.

NO	SPECIES	LOCAL NAME	COMMON NAME	COLLECTED	UNCOLLECTED
1	*Notopterus notopterus*	Belida	Knife fish	√	
2	*Anguilla* sp	Sidat	?	√	
3	*Barbichthys laevis*	Brukung	Barb	√	
4	*Barbonymus* spp	Salap	Barb	√	
5	*Barbonymus schwanenfeldii*	Salap empam	Barb		√
6	*Cosmocheilus falcifer*	Pahat	?		√
7	*Crossocheilus oblongus*	Puyau Kuncup	?		√
8	*Cyclocheilichthys apogon*	Pras	?	√	
9	*C. armatus*	Puyau	?	√	
10	*C. enoplos*	Repang	?		√
11	*C. jantochir*	Puyau	?		√
12	*Hampala macrolepidota*	Rungan	?		√
13	*Labeo chrysopekadion*	Puyau	?		√
14	*Labiobarbus* cf. *kuhlii*	Puyau	?	√	
15	*Leptobarbus hoevenii*	Jelawat	?	√	
16	*Macrochirichthys*	Parangan/Parang	?		√
17	*Osteochilus hasseltii*	Puyau	?	√	
18	*O. melanopleura*	Puyau merah mata	?		√
19	*O. waandersi*	Puyau	?		√
20	*O. pentalineatus*	Seluang nanga	?		√
21	*Oxygaster anomalura*	Lalang	?	√	
22	*Puntius tetrazona*	Ikan ban	?	√	
23	*Rasbora hubbsii*	Seluang	?	√	
24	*Thynnichthys thynnoides*	Kendia	?	√	
25	*Nematabramis* sp	Lalang	?		√
26	*Bagrichthys hypselopterus*	Ikan tikus	?		√
27	*Mystus micracanthus*	Baung	?	√	
28	*Mystus* sp.	Baung putih	?		√
29	*M. bimaculatus*	Kemotot	?		√
30	*Kryptopterus minor*	Lais	?		√
31	*K. macrocephalus*	Lepok	?		√
32	*K. lais*	Lais buntek	?		√
33	*Hemisilurus heterorhynchos*	Tilap	?		√
34	*Clarias batrachus*	Penang	Walking catfish		√
35	*Clarias nieuhofi*	Keli akar	?	√	
36	*Clarias teisjmani*	Keli biasa	?		√
37	*Xenentodon cancilloides*	Julung-julung	?	√	
38	*Hemirhamphodon* sp	Julung-julung	?	√	
39	*Macrognathus sp*	Layur	?	√	
40	*Pristolepis fasciata*	Tempe	?	√	
41	*Oxyeleotris marmorata*	Betoto	Marbled goby	√	
42	*Helostoma temmincki*	Biawan	Kissing gourami	√	
43	*Osphronemus goramy*	Kaloi	Giant gourami		√
44	*Anabas testudineus*	Puyu	?	√	
45	*Trichogaster trichopterus*	Sepat bintik	?	√	
46	*Trichogaster pectoralis*	Sepat siam	?	√	
47	*Channa lucius*	Kesong	Snake head	√	
48	*C. striata*	Aruan	Snake head	√	
49	*C. micropeltes*	Toman	Snake head	√	

The fishermen in Mesangat watershed recognize the big fish species, and they only want to get, to sell and eat the big species only. They use big mess size gill nets, buu and electroschoker, only 'anco' can get the small fish which usually they put it back to the water or they take it to fed the snake head or marble goby in the cage (we call it keramba).

The local peoples are very interested and don't know the name of the small fish which we collected surround the Mesangat rivers, they have never seen this fish before.

Figure 2. The small fishes around Mesangat river: a. *Nemacheilus cf spiniferus; b. Trichopsis vittata; c. Pangio* sp*; Rasbora* sp*; d. Betta* sp and *f. Kottelatlimia* sp.

There are three interesting species *Pangio* sp*; Rasbora* sp and *Kottelatlimia* sp. which need to study further:

- The color pattern of *Pangio* sp is different with the pattern of *P. agma* groups (*P. kuhlii,P. cuneovirgata, P. semicincta* and *P. superba*).
- The species of *Kottelatlimia sp.* need to compare with the other species *K. pristes* and *K. hyporhynchos.*
- The species of *Rasbora* sp need to compare with *R. caudimaculata* and *R. trilineata*. All of the three species have black spots on lobes of the caudal fin, but the spot of *Rasbora* sp is bigger & oval, while in *R caudimaculata* the black spots at the tips of the lobes of caudal fins, while in *R. trilineata* the spot is like thin stripes. The body of *Rasbora* sp is much thinny and slender than *R caudimaculata* and *R. trilineata*, those two species have bigger & higher body depth.

Mesangat swamp is very important for the continuation of the fish species in District Muara Ancalong, this area play a role as the spawning and feeding ground and also as a diversity fish stock. The fishermen can collect the fish in a very limited area in the Mesangat swamp, especially in the dry season, the submerge plants covered the area, only small uncovered which the boat can pass the area. Sometimes the fishesrmen operated the boat as a sky driver, using two long stick, then push it so tht the boat can move forward. The adult and the fry of the fish species hide in between the plantation.

All of the fish species deposit at Ichthyology laboratory, Zoology Division of Research Center for Biology and registered in 176 lots of MZB catalog number, from MZB 18048 -18224 (Table 3).

Tabel 3. The fish species and number of fish collection from Mesangat watershed and deposit at Ichthyology Laboratory, Zoology Division of Research Center for Biology under MZB catalog numbers:

NO	SPECIES	MZB	EXS	Date	SITES
1	*Notopterus notopterus*	18048	2	6-Oct	S. Mesangat Ulu
2	*Channa lucius*	18049	2		
3	***Macrognathus sp***	18050	2		
4	*Ompok bimaculatus*	18051	2		
5	*Xenentodon canciloides*	18052	1		
6	*Oxyeleotris marmorata*	18053	1		
7	*Trichogaster pectoralis*	18054	1		
8	*Osteochilus hasseltii*	18055	9		
9	*Oxygaster cf. anomalura*	18056	18		
10	*Helostoma temmincki*	18057	1		
11	*Anabas testudineus*	18058	1		
12	*Puntioplites waandersi*	18059	2		
13	*Parachela hypophthalmus*	18060	4		
14	*Puntius* sp.	18061	1		
15	*Labiobarbus* cf. *kuhlii*	18062	6		
16	*Trichopsis vittata*	18063	2		
17	*R. cf. hubbsii*	18064	97		
18	*Thynnichthys* sp	18065	8		
19	*Cyclocheilichthys* sp	18066	4		
20	*Puntius tetrazona*	18067	9		
21	*Hemirhamphodon* sp.	18068	2		
22	*Ompok eugeneiatus*	18069	3		
23	*Mystus* cf. *nigriceps*	18070	1		
24	*Pristolepis fasciata*	18071	1	6-Oct	S. Mesangat Ulu, Anco
25	*Leptobarbus hoevenii*	18072	3	6-Oct	
26	*Cyclocheilichthys armatus*	18073	5		
27	*Puntius binotatus*	18074	1		
28	*Osteochilus hasseltii*	18075	2		
29	*Ompok eugeneiatus*	18076	11		
30	*Helostoma temmincki*	18077	3		
31	*Ompok eugeneiatus*	18078	30	6-Oct	Loa Lakum ilir 1
32	*Rasbora cf. hubbsii*	18079	58		
33	*Channa lucius*	18080	1		
34	*Puntius tetrazona*	18081	17		
35	*Cyclocheilichthys armatus*	18082	1		
36	*Hemirhamphodon* sp.	18083	9		
37	*Puntius* sp.	18084	1		
38	*Trichopsis vittata*	18085	17		
39	*Osteochilus hasseltii*	18086	23		
40	*Rasbora* sp (black spot on CF)	18087	11	6-Oct	Mesangat Ulu, EF
41	*Rasbora* sp (black spot on CF)	18088	14	6-Oct-09	Loa Lakum ilir 1
42	*Oxygaster cf. anomalura (juv)*	18089	1		
43	*Betta sp1*	18090	4		
44	*Hemibagrus nemurus*	18091	1		

45	*Ompok bomaculatus*	18092	1		
46	*Labiobarbus* cf. *kuhlii*	18093	3		
47	*Mystus micracanthus*	18094	3		
48	*Leptobarbus hoevenii*	18095	1	7-Oct	Loa Lakum 2, depan
49	*Oxygaster cf. anomalura*	18096	1		rakit Pak Yus
50	*Puntius tetrazona*	18097	3		
51	*Rasbora cf. hubbsii*	18098	2		
52	*Thynnichthys cf thynnoides*	18099	6		
53	*Betta sp1*	18100	1		
54	*C. micropeltes (juv)*	18101	1		
55	*Trichopsis vittata*	18102	2		
56	*Hemirhamphodon* sp.	18103	2		
57	*Ompok bimaculatus*	18104	1		
58	*O.* cf. *microcephalus*	18105	5	7-Oct	Loa Lakum 3
59	*Labiobarbus* cf. *kuhlii*	18106	1		
60	*Puntius* sp.	18107	2		
61	*Rasbora cf. hubbsii*	18108	15		
62	*Cyclocheilichthys armatus*	18109	2		
63	*Oxygaster anomalura*	18110	7		
64	*Puntius tetrazona*	18111	2		
65	*Ompok bimaculatus*	18112	1		
66	*Betta sp1*	18113	6		
67	*Trichopsis vittata*	18114	1		
68	*Kottelatlimia* sp	18115	1		
69	*Pangio* sp.	18116	2		
70	*Hemirhamphodon* sp.	18117	6		
71	*Ompok eugeneiatus*	18118	4		
72	*Rasbora cf. hubbsii*	18119	119	7-Oct	Rawa Mesangat
73	*Rasbora* sp (black spot on CF)	18120	186		Scoup net
74	*Osteochilus hasseltii*	18121	1		
75	*Oxygaster anomalura*	18122	3		
76	*Parachela hypophthalmus*	18123	1		
77	*Hemirhamphodon* sp.	18124	1		
78	*Trichogaster trichopterus*	18125	2		
79	*Notopterus notopterus*	18126	1		
80	*Oxygaster anomalura*	18127	3	7-Oct	Sungai Mesangat
81	*Osteochilus hasseltii*	18128	4		Anco, morning collection
82	*Labiobarbus* cf. *kuhlii*	18129	8		
83	*Cyclocheilichthys armatus*	18130	15		
84	*Rasbora* sp (black spot on CF)	18131	1		
85	*Puntius tetrazona*	18132	1		
86	*Rasbora cf. hubbsii*	18133	15		
87	*Parachela hypophthalmus*	18134	1		
88	*Cyclocheilichthys apogon*	18135	7	7-Oct	Sungai Mesangat
89	*Leptobarbus hoevenii*	18136	1		Anco, night collection
90	*Labiobarbus* cf. *kuhlii*	18137	5		
91	*Puntioplites waandersi*	18138	3		
92	*Ompok bimaculatus*	18139	2		
93	*Ompok eugeneiatus*	18140	9		
94	*Mystus* cf. *nigriceps*	18141	8		
95	*Mystus micracanthus*	18142	1		
96	*Barbichthys laevis*	18143	1	7-Oct	Sungai Mesangat
97	*Cyclocheilichthys armatus*	18144	1		EF, night collection
98	*Leptobarbus hoevenii*	18145	6		

99	*Osteochilus hasseltii*	18146	8		
100	*Puntius tetrazona*	18147	3		
101	*Rasbora cf. hubbsii*	18148	1		
102	*Trichopsis vittata*	18149	1		
103	*Hemirhamphodon* sp.	18150	6		
104	*C. striata*	18151	2		
105	*Ompok eugeneiatus*	18152	2		
106	*Mystus micracanthus*	18153	3		
107	*Mystus* cf. *nigriceps*	18154	1		
108	*Oxygaster cf. anomalura*	18155	8		
109	*Pristolepis fasciata*	18156	4		
110	*Ompok bimaculatus*	18157	1		
111	*Xenentodon cancilloides*	18158	4		
112	*Puntius tetrazona*	18159	13	8-Oct	Loa Sangkal, tributary of
113	*Cyclocheilichthys armatus*	18160	3		Mesangat river
114	*Puntius* sp.	18161	3		
115	*Rasbora cf. hubbsii*	18162	14		
116	*Parachela hypophthalmus*	18163	1		
117	*Oxygaster anomalura*	18164	3		
118	*O.* cf. *microcephalus*	18165	10		
119	*Osteochilus hasseltii*	18166	10		
120	*Labiobarbus* cf. *kuhlii*	18167	2		
121	*Pangio* sp.	18168	8		
122	*Mystus micracanthus*	18169	10		
123	*Mystus* cf. *nigriceps*	18170	2		
124	*Ompok bimaculatus*	18171	2		
125	*Hemirhamphodon* sp.	18172	10		
126	*Pristolepis fasciata*	18173	1		
127	*Trichopsis vittata*	18174	2		
128	*Betta sp1*	18175	11		
129	*Osteochilus hasseltii*	18176	4	8-Oct	Loa Beloma, tributary of
130	*O.* cf. *microcephalus*	18177	17		Mesangat river
131	*Rasbora cf. hubbsii*	18178	3		
131	*Rasbora* sp (black spot on CF)	18179	2		
132	*Puntius tetrazona*	18180	5		
133	*Oxygaster anomalura*	18181	1		
134	*Labiobarbus* cf. *kuhlii*	18182	1		
135	*Mystus micracanthus*	18183	1		
136	*Ompok eugeneiatus*	18184	1		
137	*Ompok bimaculatus*	18185	2		
138	*Hemirhamphodon* sp.	18186	27		
139	*Pristolepis fasciata*	18187	1		
140	*Trichopsis vittata*	18188	7		
141	*Kottelatlimia* sp	18189	1		
142	*Osteochilus hasseltii*	18190	8	8-Oct	Loa Toh, tributary of
143	*O.* cf. *microcephalus*	18191	3		Mesangat river
144	*Oxygaster anomalura*	18192	2		
145	*Labiobarbus* cf. *kuhlii*	18193	3		
146	*Puntius* sp.	18194	1		
147	*Thynnichthys cf thynnoides*	18195	3		
148	*Rasbora cf. hubbsii*	18196	42		
149	*Homaloptera sp* juv.	18197	1		
150	*Nemacheilus spiniferus*	18198	1		
151	*Pangio* sp.	18199	32		

152	*Mystus micracanthus*	18200	2		
153	*Mystus* cf. *nigriceps*	18201	2		
154	*Anguilla cf bicolor*	18202	1		
155	*Trichogaster trichopterus*	18203	1		
156	*Betta sp*	18204	3		
157	*Cyclocheilichthys armatus*	18205	1	9-Oct	Long Ngetek, trib of
158	*Osteochilus hasseltii*	18206	1		Kelinjau river
159	*Oxygaster cf. anomalura*	18207	3		
160	*Rasbora cf. hubbsii*	18208	43		
161	*Kottelatlimia* sp	18209	4		
162	*Pangio* sp.	18210	11		
163	*Hemirhamphodon* sp.	18211	2		
164	*Ompok eugeneiatus*	18212	18		
165	*Mystus* cf. *nigriceps*	18213	1		
166	*Barbonymus* cf. *gonionotus*	18214	1	9-Oct	Mesangat Ilir, Desa
167	*Cyclocheilichthys armatus*	18215	1		Kelinjau, Kecamatan
168	*Parachela hypophthalmus*	18216	4		Muara Ancalong
169	*Thynnichthys cf thynnoides*	18217	3		
170	*Osteochilus hasseltii*	18218	1		
171	*Ompok eugeneiatus*	18219	3		
172	*Rasbora cf. hubbsii*	18220	55		
173	*Hemibagrus* cf. *nemurus*	18221	1	9-Oct	Mesangat river
174	*Xenentodon cancilloides*	18222	1		P Yus collection
175	*Ompok bimaculatus*	18223	1		
176	*Clarias nieuhofi*	18224	1		

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References

Brittan, M.R. 1954. A revision of the Indo-Malayan freshwater fish genus *Rasbora*. Monogr. Inst. Sci. Tech. Manila 3: 1-224.

Hadiaty, R.K. and DJ Siebert, 1998. Two new species of Osteochilus (Teleostei: Cyprinidae) from Sungai Lembang, Suag Balimbing Research Station, Gunung Leuser National Park, Aceh, Nordwestern Sumatra. Revue Franchaise d'aquariologie herpetologie Journal., 25 (1998), 1-2, 22 juin 1998**.**

Hadiaty, RK and DJ Siebert, 2001. *Nemacheilus tuberigum*, a new species of loach (Teleostei: Balitoridae) from Aceh, northwestern Sumatra, Indonesia. Bull. Nat. Hist. Mus. Lond. (Zool.) 67(2): 183-189.

Inger, R. & Chin, P.K. 1962. The freshwater fishes of North Borneo. Fieldiana Zoology 45: 1-268.

Karnasuta, J. 1993. Systematic revision of southeastern Asiatic cyprinid fish genus Osteochilus with description of two new species and a new subspecies. Kasetsart Univ. Fish. Res. Bull. No.19.

Kottelat, M. 1984. Revision of the Indonesian and Malaysian noemacheiline loaches. Jap. Journal of Ichthyol. 35.

Kottelat, M. 1990. Indochinese nemacheilines: A revision of nemacheiline loaches (Pisces: Cypriniformes) of Thailand, Burma, Laos, Cambodia and southern Vietnam. Verlag Dr. Friedrich Pfeil, Muenchen.

Kottelat, M. 1994. The fishes of the Mahakam River, East Borneo, an example of the limitations of zoogeographic analysis and the need for the extensive fish survey in Indonesia. Tropical Biodiversity 2(3): 401-436.

Kottelat, M. 1995. Four new species from Kapuas river (Teleostei*:* Cyprinidae & Belontiidae), Indonesia.The Raff. Bull. of Zool. 43(1):51-64.

Kottelat, M. 1999. Nomenclature of the genera *Barbodes, Cyclocheilichthya Rasbora* and *Chonerhinos* (Teleostei:Cyprinidae and Tetraodontidae), with comments in the definition of the first reviser. The Raffles Bulletin of Zoology 47(2): 591-600.

Kottelat, M; T. Whitten; S.N. Kartikasari & S Wirjoatmodjo, 1993. Freshwater fishes of Western Indonesia & Sulawesi. Periplus Edition, EMDI Project.

Kottelat, M. & K.K.P. Lim, 1993. A review of the eel-loaches of the genus *Pangio* (Teleostei: Cobitidae) from the Malay Peninsula, with descriptions of six new species. Raffles Bull. Of Zool 41(2): 203-249.

Kottelat, M. & Whitten, A.J. 1996. Freshwater fishes of western Indonesia and Sulawesi: addition and corrections. Periplus Edition. Jakarta.

Nelson, J. 1994. Fishes of the world. John Wiley & Sons, Inc. New York.

Ng, H.H. 1999. Two new species of catfishes of genus *Clarias* from Borneo (Teleostei: Clariidae). The Raff. Bull. Of Zool. 1999 47(1): 17-32.

Ng, H.H. & C. J. Ferraris, 2000. A review of the genus *Hemibagrus* in Southern Asia, with description of two new species. Proc. Of the Calif. Acad. Of Sciences 52(11): 125-142.

Roberts, T.R. 1989. Freshwater fishes of western Borneo. Calif. Acad. Sci. Books. California.

Roberts, T.R. 1993. Systematic revision of the southeast asian cyprinid fish genus *Labiobarbus* (Teleostei: Cyprinidae). The Raffles Bulletin of Zoology 41(2): 315-329.

Sudarto, 2003. Systematic revision and phylogenetic relationship among populations of clariid species in Southeast Asia. Abstract of Dissertation. FAMIPA, University of Indonesia, Depok.

Tan, H.H. & M. Kottelat. 1998. Two new species of *Betta* (Teleostei: Osphronemidae) From the Kapuas Basin, Kalimantan Barat, Borneo. The Raff. Bull. Of Zool. 46(1): 41-52.

Tan, H. H. & P. K. L. Ng, 2006. Six new species of *Betta* (Teleostei: Osphronemidae) from Borneo. Ichthyological Exploration of Freshwaters, 17 (2): 97-114.

Weber, M & L.F. de Beaufort. 1916. The fishes of the Indo-Australian Archipelago. III. Ostariophysi; II. Cyprinoidea, Apodes, Synbranchi, Leiden, E J. Brill Ltd.

Weber, M & L.F. de Beaufort. 1965. The fishes of the Indo-Australian Archipelago. II. Malacopterygii, Myctophoidea, Ostariophysi: Siluroidea. Leiden, E J. Brill Ltd.